SIMATS ENGINEERING
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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-6

Experiment 6: Vacuum Cleaner Problem

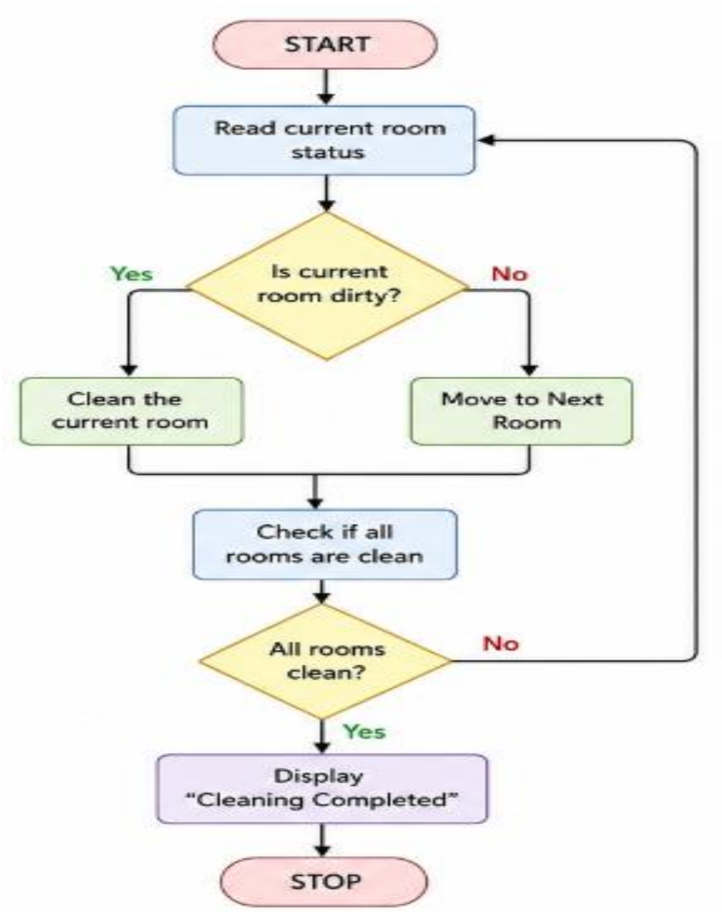
Aim

To implement the Vacuum Cleaner problem using Python.

Objective

To clean all dirty rooms by allowing the vacuum agent to move intelligently between rooms.

Flow chart



Algorithm

1. Start the vacuum in one room.
2. Check whether the current room is dirty.
3. If dirty, clean it.
4. Move to the next room.
5. Repeat until all rooms are clean.
6. Stop the program.

Code

```
rooms = ['Dirty', 'Dirty']
for i in range(len(rooms)):
    if rooms[i] == 'Dirty':
        print("Cleaning Room", i + 1)
        rooms[i] = 'Clean'


print("All Rooms Clean")
```

Output

Result

The vacuum agent cleaned all dirty rooms successfully.

Conclusion

The Vacuum Cleaner agent completed the cleaning process efficiently.